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Review Article

Electrical stimulation of neural tissue to evoke behavioral responses

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Abstract

This review yields numerous conclusions. (1) Both unit recording and behavioral studies find that current activates neurons (i.e., cell bodies and axons) directly according to the square of the distance between the electrode and the neuron, and that the excitability of neurons can vary between 100 and 4000 $\mu A/mm^2$ using a 0.2-ms cathodal pulse duration. (2) Currents as low as 10 μA, which is considered within the range of currents typically used during micro-stimulation, activate from a few tenths to several thousands of cell bodies in the cat motor cortex directly depending on their excitability; this indicates that even low currents activate more than a few neurons. (3) Electrode tip size has no effect on the current density — or effective current spread — at far field, but tip size limits the current-density generated at near field. (4) To minimize neuronal damage, the electrode should be discharged after each pulse and the pulse duration should not exceed the chronaxie of the stimulated tissue. (5) The amount of current needed to evoke behavioral responses depends not only on the excitability of the stimulated substrate but also on the type of behavior being studied.

Keywords: Electrical stimulation; Current-distance estimate; Current density; Neuronal packing density; Tissue damage; Cerebral cortex; Motor behavior

1. Introduction

The last comprehensive review written on electrical stimulation was done by Ranck in 1975. Since this time there has been much new information about the current-distance estimates of neurons, the effect of tip size on the activation of neurons, and tissue damage due to stimulation. The goal of this review is to discuss: (1) how far from the electrode tip current of a particular pulse duration activates neural tissue directly, (2) what is the effect of current density at the electrode tip on the type of neurons activated, (3) how can tissue damage be minimized during electrical stimulation, and finally, (4) how much current is required to evoke behavioral responses. These issues should be of paramount importance to anyone using electrical stimulation to study brain function.

2. Current-distance relationship

2.1. Current-distance estimates using unit recording

It is generally accepted that the amount of current injected through an electrode to activate a neuron (i.e., cell body or axon) directly is proportional the square of the distance between the neuron and the electrode tip (Asanuma and Sakata, 1967; Stoney et al., 1968; Jankowska and Roberts, 1972; Akaike et al., 1973; Roberts and Smith, 1973; Andersen et al., 1975; Gustaffson and Jankowska, 1976; Bagshaw and Evans, 1976; Shinoda et al., 1976, 1979; Marcus et al., 1979; Hentall et al., 1984a; Yeomans et al., 1988), although some studies have suggested that this relationship ranges from linear (Abzug et al., 1974) to cubic depending on the distance from the Node of Ranvier (BeMent and Ranck, 1969a, b). The conclusion that the current-distance relationship is squared is based on studies of cortical and corticospinal neurons of primates and cats (Asanuma and Sakata, 1967; Stoney et al., 1968; Andersen et al., 1975; Shinoda et al., 1976, 1979; Marcus et al., 1979), dopaminergic fibers of the medial forebrain bundle of rats (Yeomans et al., 1988), reticulospinal cells of rats (Hentall et al., 1984a), vestibulospinal and spinocerebellar fibers of rabbits and cats (Jankowska and Roberts, 1972; Akaike et al., 1973; Roberts and Smith, 1973), cell bodies of cat motoneurons (Gustaffson and Jankowska, 1976), and finally axons of spinal interneurons of cats (Jankowska and Roberts, 1972).

The first detailed study conducted on current-distance estimates of cortical neurons was performed on pyramidal cells originating from the cat motor cortex (Stoney et al., 1968). Two tungsten electrodes were inserted into the

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motor cortex, one of which was used to record from a pyramidal cell and the other of which was used to stimulate that cell (Fig. 1). A pyramidal cell was identified by antidromic activation from the pons. To activate a pyramidal cell from the cortex, a 0.2-ms cathodal pulse was delivered through the stimulating electrode. The current that evoked an action potential from the cell on 50% of stimulation trials was the threshold current of that cell at a particular electrode depth. It was found that this threshold changed with the square of the distance between the electrode and the location of minimal threshold along the electrode trajectory (Fig. 1, electrode position *b*), for stimulations above (Fig. 1, electrode position *a*) and below (Fig. 1, electrode position *c*) this location. The threshold for a particular depth, less the minimal threshold, was plotted as a function of the distance squared. The slope of this function equaled the current-distance constant (K) of the cell (Fig. 2A).

The current-distance constant ranged from 272 to 3460 $\mu A/mm^2$ for the 12 cell bodies studied. This constant was correlated ($R = 0.67$, $P < 0.02$) with the latency between the delivery of the cathodal pulse and the time at which an action potential was evoked antidromically from the pons. Consistent with this finding is the finding that this constant was negatively correlated with the conduction velocity of a neuron's axon (Hentall et al., 1984a), such that, the greater the current distance constant, the lower the conduction velocity (Jankowska and Roberts, 1972; Roberts and Smith, 1973). Therefore, the size of a neuron's axon, its conduction speed, and whether it is myelinated or non-myelinated is related to the current-distance constant.

In comparison with cell bodies of the pyramidal tract, the current-distance constant of axons of the pyramidal tract ranged from 155 to 867 $\mu A/mm^2$ when determined using a 0.3-ms pulse duration of current (Shinoda et al., 1976). After these values were adjusted to a 0.2-ms pulse duration of current by multiplying by 1.2 (1.2 times as much current, on average, was required to activate the axons using a 0.2-ms pulse as compared with a 0.3-ms pulse) the values ranged from 186 to 1040 $\mu A/mm^2$. The current-distance constant can range anywhere from 100 to 4000 $\mu A/mm^2$ depending on the neural element being activated (Table 1).

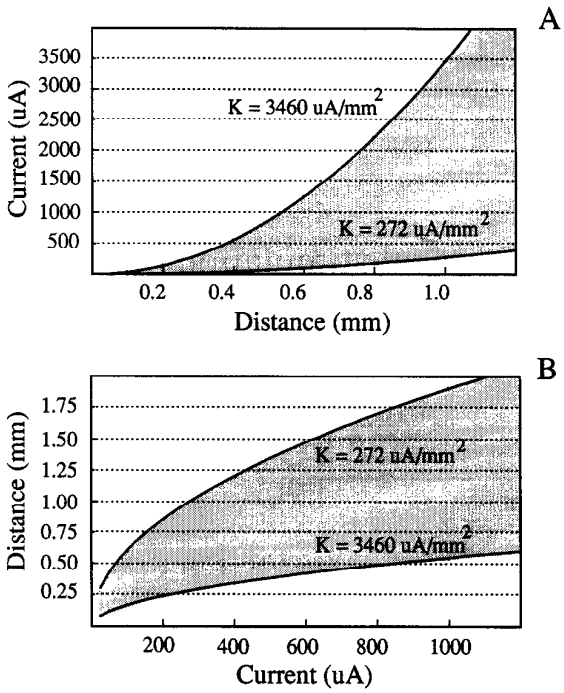


Fig. 2. A: Current is plotted as a function of distance using the current-distance relationship, $Current = K\ (Distance)^2$, where K is the current-distance constant and where *Distance* is the maximal distance a neuron can be from the electrode tip to be activated by a 0.2-ms cathodal pulse for the specified amount of current. Data are derived from Stoney et al. (1968). B: The inverse relationship of the plot in A is shown, namely: distance is plotted against current where $Distance = (Current/K)^{1/2}$. A and B: The gray area represents a range of current-distance constants, 272–3460 $\mu A/mm^2$.

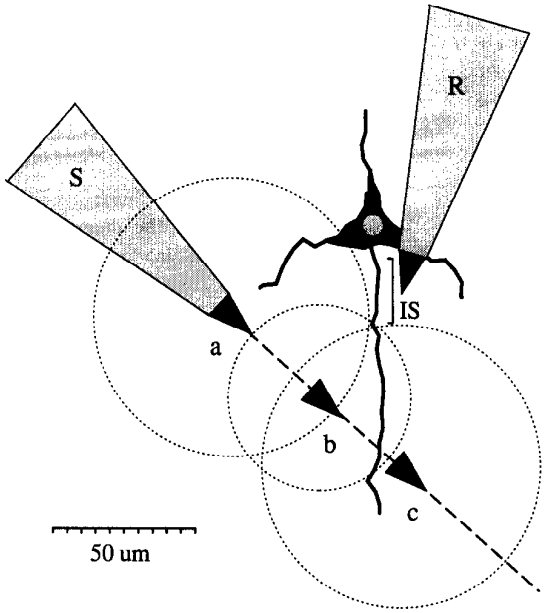


Fig. 1. The method used by Stoney et al. (1968) to determine the current-distance relationship for neurons is illustrated. The recording electrode (R) and stimulating electrode (S), both situated next to a pyramidal tract cell, are depicted to scale. The exposed microelectrode tips, shown as a black triangle, were constructed to have a diameter of 10 μm and a length of 15 μm. Each dotted circle represents the field of effective stimulation produced by 1 cathodal pulse (modelled here as a sphere) that activates the initial segment (IS) of the neuron. The initial segment is the lowest-current-threshold site at the cell body (Gustaffson and Jankowska, 1976). Initial segments of pyramidal tract neurons start at the cell body and project for 30–55 μm (Sloper and Powell, 1979b); they are typically oriented toward the white matter (Sloper et al., 1979). Each field of effective stimulation is centered on a different electrode tip position (a–c) indicating the path of the stimulating electrode. Electrode tip *b* is located at the lowest-threshold locus for current as indicated by the smaller field of stimulation. A scale bar is shown.

2.2. Current-distance estimates using behavioral methods

Several investigators have used behavioral methods to estimate how far from the electrode tip current activates neural tissue mediating behaviors such as eating (Olds, 1958), self-stimulation (Wise, 1972; Fouriezos and Wise, 1984; Milner and Laferriere, 1986), and circling behavior (Yeomans et al., 1984, 1986). The method used by Fouriezos and Wise (1984) and further developed by Yeomans et al. (1986) is now discussed.

Fouriezos and Wise (1984) determined quantitatively the current-distance relationship of neurons mediating self-stimulation in rats. Two electrodes were positioned in the medial forebrain bundle such that a line between the electrode tips was transverse to the fibers of the bundle. Current in trains of pulses (0.1-ms cathodal pulses applied for 500 ms) was delivered through each electrode (Fig. 3) following the depression of a bar by a rat. It was known a priori that the rate of bar pressing during self-stimulation increased monotonically with the frequency of stimulation pulses (Yeomans, 1975). Fouriezos and Wise delivered pairs of stimulation pulses such that the first pulse was delivered through one electrode and the second pulse was delivered through the other electrode. The second pulse followed the first pulse by an interval that was either less than or greater than the refractory period of the directly stimulated axons. If the stimulation fields were not overlapping (Fig. 3A), then both fields contributed to maintain a fixed rate of bar pressing and the frequency threshold to evoke this fixed rate was the same for short and long interpulse intervals. If, however, the stimulation fields were overlapping (Fig. 3B), then for shorter interpulse intervals a higher frequency of pulse pairs was needed to produce a fixed rate of bar pressing. This increase in frequency threshold at short interpulse intervals was assumed to be due to axonal refractoriness caused by the overlapping stimulation fields (Yeomans, 1979). In one rat whose electrodes were spaced by 0.35 mm, currents above 160 μA produced an abrupt increase in frequency threshold at short interpulse intervals. So in this case, a current of 160 μA activated axons 0.175 mm (0.35 mm/2) from the electrode tip. Assuming a squared current-distance relationship, Fouriezos and Wise calculated a minimum current-distance constant of 1300 $\mu A/mm^2$.

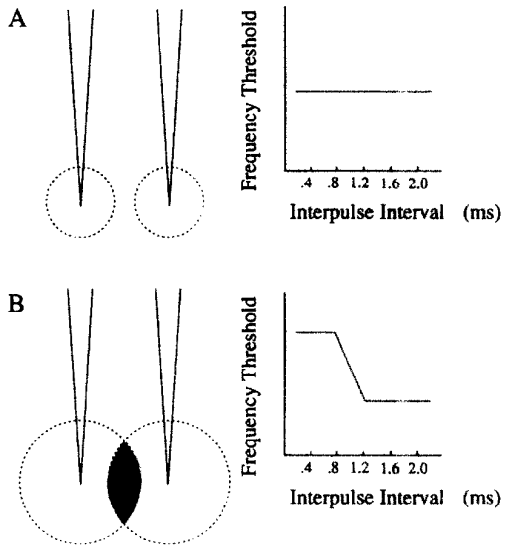


Fig. 3. A: A schematic to the left shows a pair of electrodes that have been positioned in a fiber bundle, the axons of which run through the plane of the figure. The circles about each electrode tip represent the spread of effective current. The current delivered through the electrodes is not overlapping as illustrated by the non-overlapping circles. To the right, the frequency threshold to evoke a fixed amount of behavioral responding (e.g., number of bar press or lateral body turns over time) is expressed as a function of interpulse interval when the current fields are not overlapping. B: As in A, a pair of electrodes are illustrated on the left. The current delivered through the electrodes is overlapping as indicated by the intersecting circles (black region). To the right, the frequency threshold to evoke a fixed amount of behavioral responding is express as a function of interpulse interval when the current fields are overlapping. Here the frequency threshold drops for long interpulse intervals.

Table 1
Listed according to author are various neuronal elements and their range of current–distance constants ($\mu A/mm^2$) determined using a particular pulse duration (ms). When a strength–duration curve was provide and when pulse durations other than 0.2 ms were used, the K values were adjusted (*). In case 2, the K values were multiplied by 1.2, since 1.2 times as much current was required to active the elements using a 0.2-ms pulse as compared with a 0.3-ms pulse. In case 4, the values were multiplied by 2.0, since 2.0 times as much current was required to active the elements using a 0.2-ms pulse as compared with a 0.5-ms pulse. All values were determined using metal stimulating electrodes because it has been shown that elements are more excitable using micropipettes than metal electrodes (West and Wolstencroft, 1983). Behavioral responses are generally evoked using metal electrodes

Author	Element	K ($\mu A/mm^2$)	Duration (ms)
Stoney et al. (1968)	Pyramidal tract cell bodies of cat	272–3460	0.2
Shinoda et al. (1976)	Pyramidal tract axons of cat	155–867	0.3
		186–1040	0.2 *
Marcus et al. (1979)	Pyramidal tract axons of monkey	178–667	0.25
Hentall et al. (1984a)	Reticulospinal cell bodies of rat	100–3200	0.2
Yeomans et al. (1988)	Nigral dopaminergic axons of rat	900–2000	0.5
		1800–4000	0.2 *

Yeomans et al. (1986) conducted a similar experiment as Fouriezos and Wise, but instead of assuming that axons within a stimulated bundle have uniform excitability, they estimated the distribution of current-distance constants for the axons. A pair of electrodes were implanted in the brainstem, stimulation of which evokes circling behavior. The electrodes were oriented transverse to the longitudinal fiber bundle that mediated circling (Fig. 3). The rate of circling evoked from the brainstem varied with the fre-

quency of pulses. The parameters used were 0.1-ms cathodal pulses delivered for 10 s. The current-distance constant was computed for different degrees of stimulation-field overlap. When the stimulation fields begin to overlap (at short interpulse intervals), the fibers exhibiting refractoriness should be those with low current-distance constants. When the stimulation fields are overlapped considerably, the fibers exhibiting refractoriness should include those with low as well as high current-distance constants. Increasing the degree of overlap between the stimulation fields increased the degree of refractoriness of the fiber bundle as evidenced by the increase in frequency threshold to evoke a fixed number of body turns (i.e., 3 body turns over 10 s). By comparing the pattern of increase in frequency threshold to that modelled for neurons having different current-distance constants, Yeomans et al. were able to estimate the range of current-distance constants of fibers residing in the directly stimulated fiber bundle (see Yeomans et al. (1986) for details). This analysis was also used to determine how well the effective current spread fitted a squared relationship.

Yeomans et al. found that the current-distance constants for brainstem axons mediating circling behavior in rats ranged from 200 to 4200 $\mu A/mm^2$ for a 0.1-ms pulse duration. This range overlapped with the wide range of current-distance constants exhibited by medullospinal neurons of rats which ranged from 134 to 4288 $\mu A/mm^2$. [1] Furthermore, the current-spread data of Yeomans et al. fitted a squared relationship best indicating that current activates neurons according to the square of the distance between an electrode and neuron (Asanuma and Sakata, 1967; Stoney et al., 1968; Jankowska and Roberts, 1972; Akaike et al., 1973; Roberts and Smith, 1973; Andersen et al., 1975; Gustaffson and Jankowska, 1976; Bagshaw and Evans, 1976; Shinoda et al., 1976, 1979; Marcus et al., 1979; Hentall et al., 1984a; Yeomans et al., 1988).

It is possible that the change in frequency threshold observed by Yeomans et al. for the higher-current condition was due to the synaptic activation of nerve fibers residing outside the region of stimulation. There is much evidence, however, suggesting that the change in frequency threshold is due to direct activation of axons residing about the electrode tip (Yeomans, 1975, 1979, 1990; Shizgal et al., 1980; Miliaressis, 1981; Yeomans and Linney, 1985; Yeomans and Tehovnik, 1988). The stimulation of pons to evoke circling behavior has been used to show that the stimulated elements exhibited axonal properties (Miliaressis, 1981; Yeomans and Linney, 1985; Tehovnik and Yeomans, 1986, 1987; Yeomans and Tehovnik, 1988; Yeomans, 1990, 1995): namely, local potential summation effects, relative and absolute refractory period effects, supernormal period effects, and axonal conduction velocity estimates that agreed with the conduction velocities of axons known to reside within the stimulated sites.

In conclusion, current-distance estimates using behavioral methods yielded results that were similar to those obtained with unit recording. Specifically, the directly stimulated axons mediating behavioral responses exhibited a broad range of current-distance constants similar to those exhibited by individual neurons studied with unit recording, and the current activated axons according to the square of the distance between the electrode tip and nerve fiber.

2.3. Current-distance estimates for cortical neurons involved in evoking saccadic eye movements

A behavior studied extensively using electrical stimulation are eye movements. Such movements are readily evoked by delivering trains of electrical pulses to particular regions of the cerebral cortex (Wagman, 1964). A wide range of currents (5–3000 μA) and pulse durations (0.1–1.0 ms) of cathodal pulses have been found to evoke saccadic eye movements from the cortices of unanesthetized, behaving monkeys (e.g., Robinson and Fuchs, 1969; Schiller, 1977; Goldberg et al., 1986; Tehovnik and Lee, 1993). Currents as high as 3000 μA (at 0.2-ms duration) have been used to evoke saccades from the occipital cortex (Schiller, 1977) and currents as low as 10 μA (at 0.25-ms duration) have been sufficient for evoking saccades from the frontal eye fields (Bruce et al., 1985).

Using the current-distance constants of Stoney et al. (1968), the effective current spread can be estimated for the different currents and pulse durations used to evoke saccades from monkey cortex. Here it is assumed that the directly stimulated elements of cortex are similar to the pyramidal cells of motor cortex and that the current-distance constants of monkey pyramidal cells are similar to those of cat pyramidal cells. Fig. 2B illustrates the distance of effective current spread plotted against current for a 0.2-ms cathodal pulse. In this case, the amount of current necessary to activate a neuron 1 mm away from the electrode tip would be 272 μA for a low- and 3460 μA for a high-threshold neuron.

Since different pulse durations of cathodal pulses have been used to evoke saccades from cortex, the effective current spread for various pulse durations is shown for low- and high-threshold neurons (Fig. 4). The current threshold to evoke an action potential from a pyramidal cell decreased when the pulse duration increased from 0.1 to 0.5 ms (Stoney et al., 1968; Asanuma et al., 1976), but for pulse durations beyond 0.5 ms the current threshold remained relatively constant. This lowest, asymptotic, cur-

[1] The actual current–distance constants (100–3200 $\mu A/mm^2$) were determined with a 0.2-ms pulse (Hentell et al., 1984a) so the values had to be adjusted to a 0.1-ms pulse duration by multiplying by 1.34, since 1.34 times as much current (on average) was required to activate the neurons using 0.1-ms pulses as compared with 0.2-ms pulses (Hentall et al., 1984b).

rent is termed the "rheobase current". Current threshold normalized to the rheobase current is plotted as a function of pulse duration for 6 pyramidal neurons (Fig. 5A). On average, 1.8 and 0.6 times as much current was required to activate neurons with a 0.1- and 0.5-ms pulse duration, respectively, when compared to the current required to activate neurons with a 0.2-ms pulse duration, the duration used for the current-distance functions of Fig. 2. By multiplying the current-distance constants for low- and high-threshold neurons by 1.8 and 0.6, respectively, current-distance functions can be computed for 0.1- and 0.5-ms duration pulses. So, for low-threshold neurons (i.e., a 272 $\mu A/mm^2$ current-distance constant at 0.2-ms pulse duration), a 1000 μA current would activate neurons 1.43, 1.92, and 2.48 mm from the electrode tip, respectively, when using pulse durations of 0.1, 0.2, and 0.5 ms. For high-threshold neurons (i.e., a 3460 $\mu A/mm^2$ current-distance constant at 0.2-ms pulse duration), a 1000 μA current would activate neurons 0.4, 0.54, and 0.69 mm from the electrode tip, respectively. Finally, for pulse durations from 0.5 to 1.0 ms, current would activate neurons at the same maximal distance from the electrode tip, since the current threshold to evoke an action potential at these durations was similar (Stoney et al., 1968; Asanuma et al., 1976).

Recall that the foregoing assumes that the directly stimulated neurons that mediate saccadic eye movements from cortex of monkeys exhibit the same excitability (i.e., have the same distribution of current-distance constants) as pyramidal cell bodies from the motor cortex of cats. It has been found that the change in current threshold to evoke

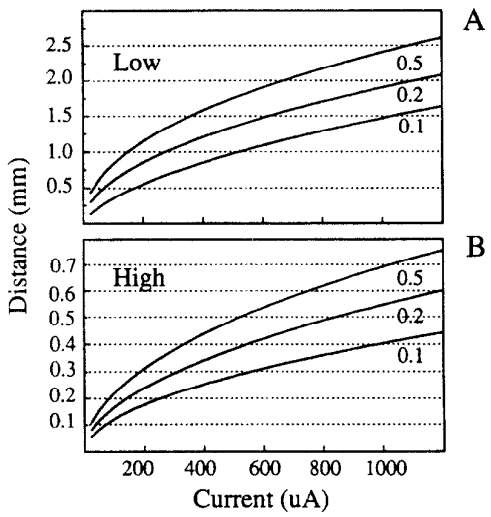


Fig. 4. Distance is plotted as a function of current for low- and high-threshold neurons using the relationship $Distance = (Current/K)^{1/2}$ (see Fig. 3). A: For low-threshold neurons, the current-distance constants are 490, 272, and 163 $\mu A/mm^2$ for 0.1-, 0.2-, and 0.5-ms pulse durations, respectively. B: For high-threshold neurons, the current-distance constant are 6228, 3460, and 2076 $\mu A/mm^2$ for 0.1-, 0.2-, and 0.5-ms pulse durations, respectively.

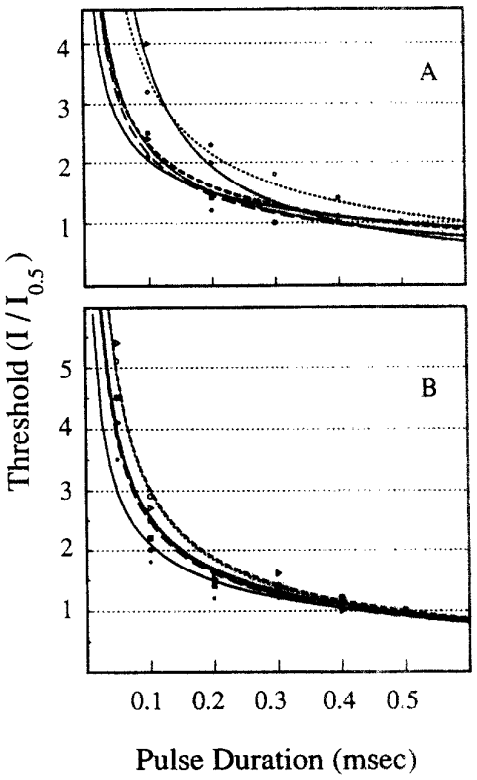


Fig. 5. A: Current threshold normalized to the rheobase current (at 0.5-ms duration) to evoke an antidromic action potential on 50% of stimulation trials is plotted as a function of pulse duration for 6 pyramidal cell units. The data are replotted from Asanuma et al. (1976) and Stoney et al. (1968) by using power functions that account for over 84% of the variability in current threshold tested at pulse durations of 0.1, 0.2, 0.3, 0.4, and 0.5 ms for each of 6 units. The chronaxies (the pulse duration at twice the rheobase current) range from 0.10 and 0.22 ms for the units shown. B: Current threshold normalized to the rheobase current (at 0.5-ms duration) to evoke a saccade on 50% of stimulation trials is plotted as a function of pulse duration for 6 stimulation sites in the dorsomedial frontal cortex of the rhesus monkey (Tehovnik and Lee, 1993). The chronaxies range from 0.1 and 0.18 ms.

saccades from the dorsomedial frontal cortex of monkeys was similar to the change in current threshold to evoke action potentials from pyramidal cells of cats for pulse durations between 0.05 and 0.5 ms (cf., Fig. 5A,B). In both cases, the chronaxie (i.e., the pulse duration at twice the rheobase current) ranged between 0.10 and 0.22 ms, suggesting that the excitability of neurons mediating stimulation-evoked saccades are similar to those of pyramidal cells of cat. Current-distance estimates for cortical neurons mediating saccades need to be determined using the behavioral methods of Yeomans et al. (1986) to further validate the assumption.

The current-distance estimates for the neurons mediating saccades pertain to direct activation only and not to indirect activation, which has also been studied for pyramidal-tract cells (Asanuma and Rosen, 1973; Jankowska et al., 1975; Asanuma et al., 1976). The degree of indirect

activation of neurons mediating saccades evoked by trains of stimulation needs to be studied.

3. Stimulation of cerebral cortex

3.1. Direct activation of neurons while stimulating cerebral cortex

The number of cells within a cylinder of constant cross-sectional area of 1 mm^2 of cortex is remarkably constant amongst different mammalian species and their different cortical areas (Bok, 1959; Rockel et al., 1980; Braitenberg and Schuz, 1991), with one exception that this number is 2.5 times greater for the striate cortex of primates (Rockel et al., 1980). Rockel et al. (1980) counted roughly 120000 neurons/mm^2 through the entire depth of the cerebral cortex of mammals (i.e., rat, cat, monkey, and man) which can vary in thickness from 1 to 3 mm depending on the species and cortical area.

The thickness of cat motor cortex is approximately 1.6 mm (Rockel et al., 1980). Therefore, the number of neurons within cat motor cortex is 75 cells/0.001 mm^3. According to estimates of Stoney et al. (1968) of cat motor cortex, the volume of tissue activated directly by a 10 μA current (at 0.2-ms duration) would be 0.029 mm^3 for low-threshold neurons ($4/3\ \pi\ (r)^3$, where $r = 0.19$ mm, the effective current spread for low-threshold neurons (272 $\mu A/mm^2$)) and 0.0005 mm^3 for high-threshold neurons ($4/3\ \pi\ (r)^3$, where $r = 0.05$ mm, the effective current spread for high-threshold neurons (3460 $\mu A/mm^2$). Since 75 cells occupy a volume of 0.001 mm^3, then a 10 μA current (at 0.2-ms duration) should activate 2175 low-threshold neurons directly (0.029 $mm^3/0.001\ mm^3 \times 75$ cells) and 38 high-threshold neurons directly (0.0005 $mm^3/0.001\ mm^3 \times 75$ cells) (Fig. 6). Accordingly, the total number of neurons activated directly by a given current is not only dependent on the amount of current used but also on the excitability of neurons.

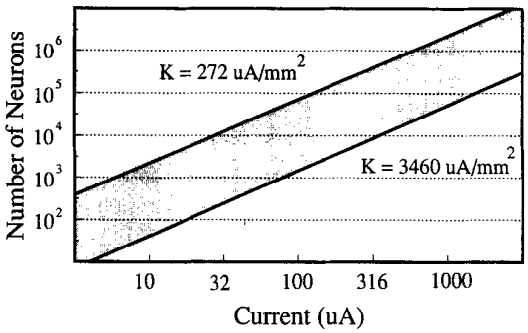


Fig. 6. The number of neurons activated is plotted as a function of current using a log-log scale. Curves for low (272 $\mu A/mm^2$) and high (3460 $\mu A/mm^2$) current-distance constants are shown. The grey region represents neurons with current-distance constants between these extremes. To generate the functions, the radius of effective current spread from the electrode tip was determined using *Radius* = (*Current*/*Current* − *Distance Constant*)$^{1/2}$, the volume of effective current spread was determined by computing the volume of a current sphere, *Volume* = $4/3\ \pi$ (*Radius*)^3, and the number of neurons activated in the sphere was determined by *Number of Neurons* = *Volume* × 75 cells/0.001 mm^3.

3.2. Transynaptic effects during cortical stimulation

Currents below 50 μA (at 0.2 ms duration) typically fail to activate the output cells of cortex directly unless the electrode is presumptively situated near the initial segment of the neuron (Jankowska et al., 1975). Asanuma and Rosen (1973) found that a 4 μA (at 0.2-ms duration) cathodal pulse delivered to layers II, III, or V of cat motor cortex monosynaptically activated neurons situated within 0.5 mm from the electrode tip, and polysynaptically activated neurons situated within 1.0 mm from the electrode tip. These transynaptic effects were always greater across the layers than along the layers. Also these effects are more marked when evoked by tetanic stimulation.

When evoking saccades during cortical or tectal stimulation, for example, it is common for trains of pulses to be delivered at or below 400 Hz (e.g., Robinson, 1972; Schiller and Stryker, 1972; Guitton et al., 1980; McIlwain, 1982; Bruce et al., 1985; Schlag and Schlag-Rey, 1987; Tehovnik et al., 1994). McIlwain (1982) stimulated the intermediate layers of the superior colliculus at 400 Hz using 4 cathodal pulses each at 30 μA for a 0.5-ms duration. He found that some 80% of the neurons situated within 1 mm of the electrode tip were activated synaptically. A 30 μA current at a 0.5-ms duration should activate neurons situated within 0.1 and 0.4 mm of the electrode tip directly, depending on their current-distance constant (i.e., 2076 $\mu A/mm^2$ for high-threshold neurons, and a 163 $\mu A/mm^2$ for low-threshold neurons, see Fig. 4). This suggests that when high frequencies are being delivered, many neurons located beyond the fringe of effective current spread are also activated, but the distance of synaptic action beyond this fringe is very dependent on the strength of synapses and on the connectivity of the directly stimulated site.

3.3. Cortical mapping

Most unit activity encountered by an advancing electrode likely occurs when the electrode is located in layers II, III and V of cortex, the laminae that contain the largest neurons, i.e., the pyramidal cells (Tombol, 1974; Sloper et al., 1979; Sloper and Powell, 1979a; Rockel et al., 1980). The lowest-current-threshold sites for evoking motor responses from cortex have been found between layers II and V (Asanuma and Sakata, 1967; Asanuma and Rosen, 1972; Andersen et al., 1975; Asanuma and Arnold, 1975). If one considers a cortex with a approximate depth of 2 mm, which is the average thickness of monkey cortex (Rockel et al., 1980), layers II and V would be situated

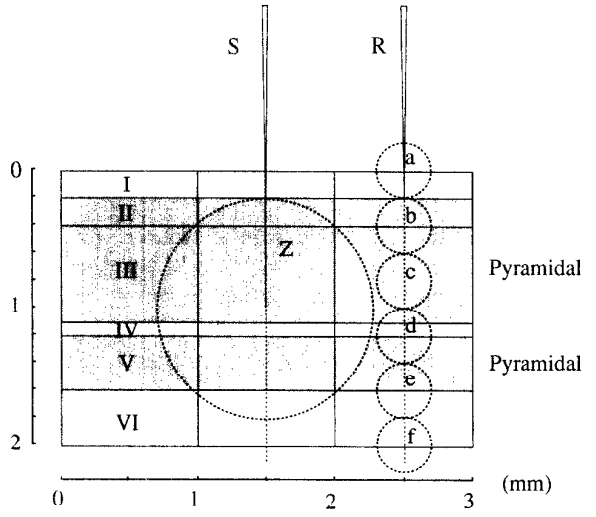


Fig. 7. A schematic is shown of layers I–VI of the cerebral cortex. Layers II, III and V, depicted in grey, contain most pyramidal cells (Sloper and Powell, 1979a). A field of unit recording (with radius of 0.2 mm) is shown as a circle at different depths (a–f) within cortex with electrode R. A field of electrical stimulation (with radius of 0.8 mm) is shown as a circle centered at 1 mm below the cortical surface. See text for details.

between 0.2 and 1.6 mm from the cortical surface (Rockel et al., 1980).

In order to record from an active neuron, an electrode must be located within 0.2 mm of that neuron: an electrode that can directly activate a unit with 10 μA (at 0.2-ms duration) or less can also be used to record from that unit (Stoney et al., 1968). The effective current-spread for a 10 μA current is 0.2 mm for low-threshold neurons (272 $\mu A/mm^2$ at 0.2-ms duration). Fig. 7 shows a recording electrode (R), located at different depths (a–f) in a 2-mm-thick cortex. At each electrode depth, a recording field is indicated by a dotted circle with radius of 0.2 mm. Assuming that the pyramidal layers contain the active cells, units should be encountered at electrode depths between 0.1 and 1.8 mm (a–e but not f). It is not unreasonable to assume that an electrode (S) would be positioned somewhere between 0.1 and 1.8 mm from the cortical surface while stimulating cortical layers, when using unit recording to guide electrode placement.

If the electrode is positioned in the centre of cortex at a 1-mm depth, for instance, then in order to activate the neurons within layers II, III and V, the effective current must spread about 0.8 mm (Fig. 7, stimulation field z). A current of at least 174 μA would be needed to activate low-threshold neurons directly (272 $\mu A/mm^2$ at 0.2-ms duration) through these layers.

Tehovnik and Lee (1993) found that a fixed current of 235 μA (adjusted to a 0.2-ms pulse duration) was optimal for describing the topographic map of the dorsomedial frontal cortex representing saccadic eye movements. Electrical stimulation was typically delivered by an electrode 1 mm below the first unit encountered in a pass (which put the electrode near the centre of the 2-mm-thick cortex). Others failed to find this topographic order apparently since they used lower currents (which varied from 3 to 150 μA at 0.2-ms duration) and since they did not control their electrode depth (Schlag and Schlag-Rey, 1987; Russo and Bruce, 1993). Hence, mapping studies should use a current that maximally activates pyramidal layers in cortex from an optimal electrode depth. This strategy increases the probability that the major output layer of cortex, i.e., layer V (Sloper et al., 1979), is activated even if the electrode tip is above this region.

Jankowska et al. (1975) demonstrated that currents less than 50 μA (at 0.2-ms duration), more often than not, did not activate the pyramidal cells of cat or monkey motor cortex directly. Yet they found that surface stimulation using currents between 200 and 1500 μA reliably acti-

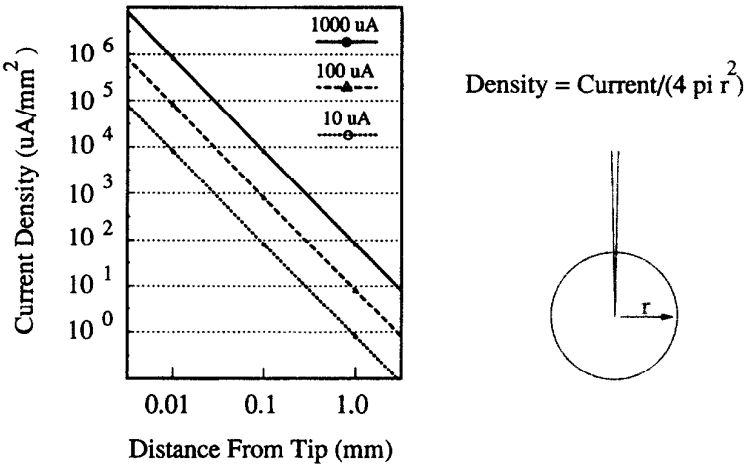


Fig. 8. Current density is plotted as a function of distance from the electrode tip for 10, 100, and 1000 μA currents. The relationship, $Density = Current/(4 \pi r^2)$, is used where r is the distance from the electrode tip in millimeters. To the left of the figure is shown an electrode passing a current with a radial spread of r.

vated these cells directly. These currents activate low-threshold neurons (272 $\mu A/mm^2$ at 0.2-ms duration) between 0.9 and 2.4 mm from the electrode tip, the range in distance that overlaps with the thickness of cat and monkey motor cortex that can range from 1 and 3 mm (Rockel et al., 1980). Jankowska et al. also found that if the stimulating electrode was advanced 200–400 μm below the site of best unit recording of a pyramidal cell, which they assumed put the electrode closer to the initial segment of that cell, the cell could be excited directly. To activate the output cells of cortex directly, currents greater than 50 μA (at 0.2-ms duration) should be used routinely, and the electrode should be positioned within the deep rather than superficial layers of cortex.

4. Current density

At far field, the current density (current divided by the surface area of a sphere) needed to evoke an action potential from a neuron is dependent only on the neuron's distance from the electrode tip and on the amount of current delivered at a particular pulse duration, and is independent of the surface area of the electrode (Fig. 8). This means, therefore, that effective current spread at far field is not affected by the size of an electrode tip.

The current threshold of a neuron has been expressed by some researchers as the current-density threshold (or current flux) which is the current-distance constant of a neuron divided by 4 π (Milner and Laferriere, 1986). For pyramidal tract neurons that have current-distance constants varying from 272 to 3460 $\mu A/mm^2$, their current-density thresholds would vary from 22 to 275 $\mu A/mm^2$. The current-density threshold is the amount of current-density needed to activate a neuron 1 mm away from the electrode tip.

For neurons situated near to the electrode tip, the current density generated at the tip (i.e., current divided by the surface area of the electrode) for a given current is limited by the size of the electrode tip. It was found that currents between 7.4 and 400 μA were required to evoke action potentials from reticulospinal neurons when using large-tipped electrodes (0.05 mm^2), whereas currents between 1.6 and 20 μA were required to evoke such potentials using small-tipped electrodes (0.0001 mm^2) (West and Wolstencroft, 1983). Likewise, Bagshaw and Evans (1976) found that more current was required to stimulate the sciatic nerve of frog when using large-tipped electrodes (0.1–0.5 mm^2) than when using small-tipped electrodes (0.07–0.09 mm^2).

Yeomans et al. (1986) found that the relationship between frequency and current (which was approximately linear) for evoking a fixed amount of circling behavior from the brainstem of rats differed according to whether the electrode tip was large (0.05 mm^2) or small (0.01 mm^2). At a given current, higher frequency thresholds were found when using large-tipped electrodes as compared with small-tipped electrodes. Furthermore, behaviorally determined refractory periods of axons mediating stimulation-evoked circling were shorter, on average, when using large-tipped electrodes (0.6 mm^2) than when using small-tipped electrodes (0.01 mm^2) (Yeomans et al., 1985).

Yeomans and colleagues have attributed these differences to the greater current density that was generated at the tip of a small electrode as compared with a large electrode. The higher current density recruited more neurons at a given current level, since fibers about the electrode tip having high-current thresholds (and longer refrac-

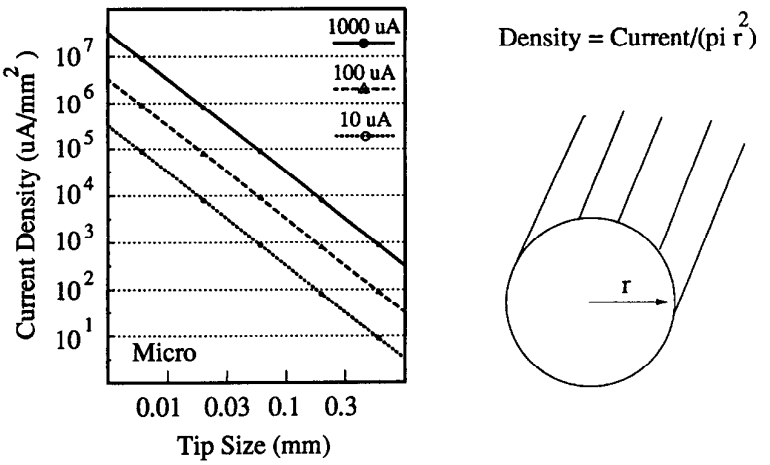


Fig. 9. Current density is plotted as a function of electrode tip size for 10, 100,and 1000 μA currents. The relationship, *Density = Current/(4 π r^2)*, is used where r is the radius (or size) of the electrode tip in millimeters. To the left of the figure is shown an electrode with radius r. 'Micro' indicates the size of a typical metal microelectrode (Stoney et al., 1968; Shinoda et al., 1979; Lemon, 1983).

tory periods) were now added to the stimulated population of neurons.

The exposed surface area of an electrode used to evoke behavioral responses can vary from 0.0001 mm^2 (e.g., Mitz and Wise, 1987) to 0.6 mm^2 (e.g., Yeomans et al., 1985). Fig. 9 shows how current density at an electrode tip varies with the radius of an electrode tip. The range of tip sizes and currents illustrated corresponds to those most often used to evoke behavioral responses from brain tissue. Since tip size affects the current-density generated at near field it is important to mention the size of one's electrode tips.

Varying the size of the electrode tip can be used to recruit neurons with different excitability. For example, if one needs to recruit fibers exhibiting high current-distance constants, such as the dopaminergic fibers from the substantia nigra (Yeomans et al., 1988), one could use small-tipped electrodes to more readily activate these fibers. This is particularly true when low currents need to be used to restrict effective current spread.

In conclusion, at far field effective current spread is not affected by electrode tip size, and in order to recruit high-threshold neurons using low currents, small-tipped electrodes should be used.

5. Tissue damage

5.1. Noxious effects of stimulation

Asanuma and Arnold (1975) studied the noxious effects of high currents. Neuronal damage from low- and high-

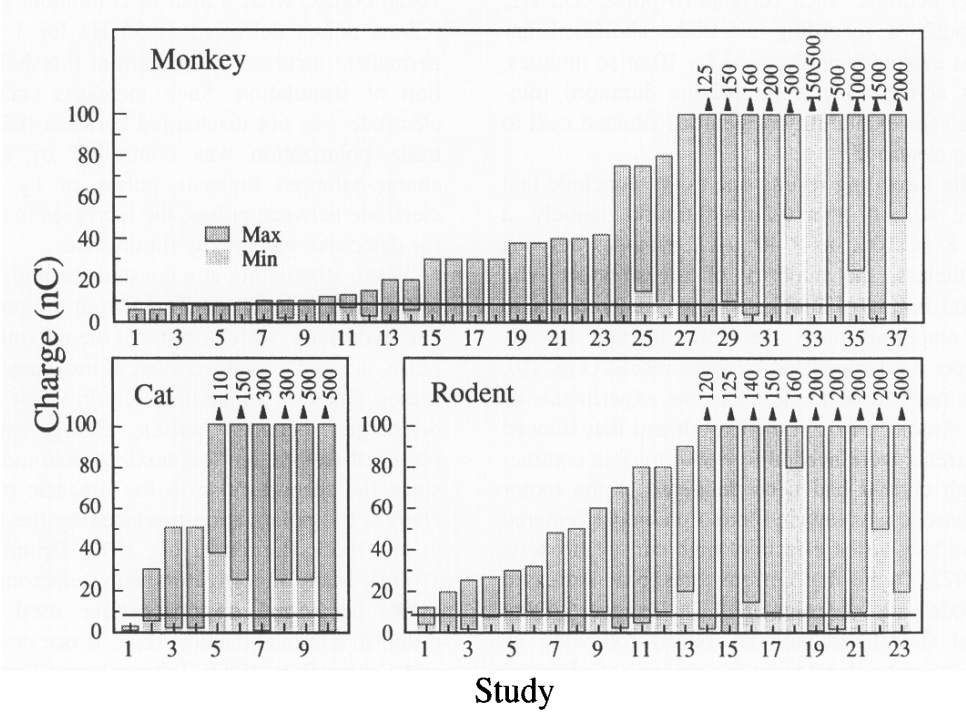


Fig. 10. The amount of charge in a cathodal pulse used to evoke behavioral responses in studies using monkeys, cats, and rodents is shown. The bars indicate the range of charges used per phase. Studies are rank ordered according to the amount of charge. In cases where the charge was greater than 100 nC, the number above the bar indicates the upper limit. In one case (Monkey, study 33), both the lower and upper limit are indicated. All values, rank ordered from low to high, are derived from the following studies. *On monkeys*: (1) Gould et al., 1986; (2) Nodo and Fujikado, 1987; (3) Tanji and Kurata, 1982; (4) Rizzolatti et al., 1990; (5) Luppino et al., 1991; (6) Alexander and Crutcher, 1990; (7) Sparks et al., 1987; (8) Sparks and Mays, 1983; (9) Macpherson et al., 1982; (10) Mitz and Godschalk, 1989; (11) Mitz and Wise, 1987; (12) Cohen et al., 1985; (13) Schlag and Schlag-Rey, 1987; (14) Parthasarathy et al., 1992; (15) Cohen and Komatsuzaki, 1972; (16) Gottlieb et al., 1993; (17) Stanton et al., 1989; (18) Russo and Bruce, 1993; (19) Goldberg et al., 1986; (20) Bruce et al., 1985; (21) Tehovnik et al., 1994; (22) Funahashi et al., 1989; (23) Shibutani et al., 1984; (24) Schiller and Stryker, 1972; (25) Kurylo and Skavenski, 1991; (26) Tehovnik and Lee, 1993; (27) Huerta and Kaas, 1990; (28) Huerta et al., 1986; (29) McElligott and Keller, 1984; (30) Schiller and Sandell, 1983; (31) Stryker and Schiller, 1975; (32) Mohler et al., 1973; (33) Ron and Robinson, 1973; (34) Robinson, 1972; (35) Marrocco, 1978; (36) Schiller, 1977; (37) Robinson and Fuchs, 1969. *On cats*: (1) Straschill and Rieger, 1973; (2) McIlwain, 1986; (3) Roucoux et al., 1980; (4) Guitton et al., 1980; (5) Syka and Radil-Weiss, 1971; (6) Guitton and Mandl, 1978; (7) Harris, 1980; (8) Schlag and Schlag-Rey, 1970; (9) Schlag and Schlag-Rey, 1971; (10) McIlwain, 1988. *On rodents*: (1) Milner and Laferriere, 1986; (2) Donoghue and Wise, 1982; (3) Neafsey and Sievert, 1982; (4) Miliaressis, 1981; (5) Castaneda et al., 1985; (6) Sabibzada et al., 1986; (7) Tehovnik and Yeomans, 1988; (8) Robinson, 1978; (9) Tehovnik, 1989; (10) McHaffie and Stein, 1982; (11) Yeomans et al., 1986; (12) Yeomans and Linney, 1985; (13) Yeomans and Buckenham, 1992; (14) Northmore et al., 1988; (15) Neafsey et al., 1986; (16) Tehovnik and Yeomans, 1986; (17) Sinnamon and Bradley, 1984; (18) Ellard and Goodale, 1986; (19) Atrens and Cobbin, 1976; (20) Yeomans et al., 1984; (21) Yeomans et al., 1985; (22) Tehovnik and Yeomans, 1987; (23) Wilcott, 1979. The black line at the bottom of each set of bar graphs depicts the 8 nC upper limit suggested by Asanuma and Arnold (1975).

current conditions were compared on stimulation-evoked muscle contractions of cat forearm muscles. An electrode was advanced into the motor cortex and low currents (< 15 μA, 17 pulses at 400 Hz, 43-ms train) were used to determined the excitable region between 0.1 and 1.5 mm below the cortical surface. Three μA were needed to evoke responses from the lowest-threshold site that was 1.1 mm below the cortical surface. The electrode was withdrawn and re-introduced using high currents (< 80 μA, 6 pulses at 500 Hz, 12-ms train). After stimulating a site with the high currents, 10 minutes later the lower currents were applied. Now the current threshold at the lowest-threshold site using the lower-current condition was 8 μA which was higher than the 3 μA current originally needed to evoke a response.

Asanuma and Arnold also found that currents above 40 μA (at 0.2-ms duration) and up to 80 μA damaged pyramidal tract neurons. Such currents (6 pulse, 333 Hz) delivered through a recording electrode abolished antidromic spikes evoked from the pons for 10 or so minutes. Thus currents above 40 μA (at 0.2-ms duration) transiently distorted the excitability of neurons situated next to the stimulating electrode.

Based on the foregoing results one might conclude that currents above 40 μA at a 0.2-ms duration, namely, a charge above 8 nC (0.2 ms × 40 μA), damages nervous tissue. Nevertheless, the majority of investigators who have stimulated brains of monkeys, cats, and rodents to evoke motor responses have used currents that far exceeded the upper limit suggested by these results (Fig. 10).

Two points need to be made about the experiments of Asanuma and Arnold. First, in the experiment that showed that higher currents were needed to evoke muscle contractions after high current had been delivered to the motor cortex, the electrode was lowered into a previously entered electrode path to study the effects of high current. Roberts and Smith (1973) found that current thresholds increased after an electrode path was re-entered. They suggested that a low-resistant fluid-filled path is created following an electrode pass, thereby decreasing the amount of effective current activating the tissue. Because such a low-resistance path may have been produced in the experiments of Asanuma and Arnold, it is unclear whether their increase in current threshold was actually due to high current or due to the re-introduction of the electrode.

Second, in the study showing that delivering trains of cathodal pulses at currents greater than 40 μA transiently disrupt antidromic responses, it was found that when cathodal currents above 40 μA (at 0.2-ms duration) were delivered through a small-tipped metal recording electrode (with surface area of 0.0007 mm^2) bubbles, likely due to hydrolysis (Brummer and Turner, 1975, 1977b; Brummer et al., 1977; Bartlett et al., 1977), were emitted from the electrode tip (Asanuma and Arnold, 1975). In none of the experiments conducted by Asanuma and Arnold was the electrodes discharged between the delivery of individual cathodal pulses. Accordingly, the transient diminution of the antidromic response was likely related to electrode polarization and hydrolysis (Brummer and Turner, 1975, 1977b; Brummer et al., 1977; Bartlett et al., 1977). Also the delivery of current through their small-tipped electrodes likely caused much corrosion at the tip (Bernstein et al., 1977; Johnson and Hench, 1977), which may have also contributed to the transient damage.

5.2. *Stimulating pulses*

Using a behavioral criterion to assess neuronal damage due to stimulation, Bartlett et al. (1977) concluded that most neuronal damage can be attributed to electrode polarization and hydrolysis, which produces a dramatic change in tissue pH as well as production of gas. They trained monkeys to detect electrical stimulation delivered to the visual cortex. After a bout of continuous stimulation (e.g., 0.2-ms pulses delivered at 50 Hz for 1 h), there was a permanent increase in the current threshold for the detection of stimulation. Such increases occurred when the electrode was not discharged between pulses. When electrode polarization was controlled by stimulating with charge-balanced biphasic pulses or by discharging the electrode between pulses, the increases in current threshold for detection was greatly diminished.

When stimulating at a constant current, each pulse adds to the voltage level remaining from the preceding pulse so that hydrolysis, which occurs at the maximal voltage of the pulse, is rapidly attained even at low currents. It is for this reason that safe stimulation requires that the electrode be discharged between pulses. Charge-balanced biphasic pulses, therefore, are less noxious than monophasic pulses, since the second pulse in the biphasic pair actively discharges the polarization produced by the first pulse (Lilly et al., 1955; Bartlett et al., 1977; Brummer and Turner, 1977a). Many investigators using electrical stimulation to evoke behavioral responses have used this method to diminish neuronal damage (e.g., Bruce et al., 1985; Schlag and Schlag-Rey, 1987; Tehovnik and Lee, 1993). Others, however, have electronically discharged their electrodes between pulses to reduce damage (e.g., Yeomans and Linney, 1985; Tehovnik and Yeomans, 1986).

Rowland et al. (1960) found that charge-balance biphasic stimulation with fewer than 20 000 nC/phase could be delivered to the cat cortex for a total of 10 C without producing an electrolytic lesion at an electrode with a 3 mm^2 exposed tip. As the charge was increased beyond 25,000 nC/phase, there was a sharp increase in the size of the lesion, such that the size increased monotonically with charge/phase. The size of the lesion was not dependent on the frequency and waveform of pulses; nor did it depend on the period between the anodal and cathodal pulses. Using more sensitive criteria for tissue damage (i.e., by noting glycogen granules, astrocytes, cytoplasmic vacuolization, and neuronophagia), Pudenz et al. (1975, 1977)

also found that the size of the lesion increased with charge/phase once a particular threshold was surpassed. In their studies, the threshold for neuronal damage was 300 nC/phase for charged-balanced biphasic pulses delivered for 36 h at 50 Hz through a 1 mm^2 electrode.

In conclusion, although transient neuronal damage is evident at charge as low as 8 nC/phase (Asanuma and Arnold, 1975), the delivery of charge-balanced biphasic pulses or monophasic pulses with interpulse electrode depolarization can greatly diminish neuronal damage. Once a particular threshold for neural damage is surpassed (e.g., 300 nC/phase: Pudenz et al., 1975, 1977) tissue damage increases as a function of charge per phase.

The minimization of charge per phase may be achieved by applying short duration pulses (Crago et al., 1974). Using pulses that are not greater than the chronaxie insures that most of the applied charge is going toward the evocation of the response. Pulse durations significantly greater than the chronaxie do not contribute significantly to the evoked response, even though they will continue to contribute to neuronal damage. The chronaxies should be known when investigating stimulation-evoked responses.

5.3. Charge density and electrode tip size

As discussed previously, electrode tip size has an effect on current density generated at the electrode tip. Pollen (1977) found that small-tipped electrodes were more detrimental to neural tissue than large-tipped electrodes. He stimulated cat motor cortex with electrodes with exposed tips varying from 0.05 to 0.8 mm^2 in surface area. The current threshold to evoke an after-discharge was about 50% lower for a 0.05 mm^2 electrode than for a 0.8 mm^2 electrode when the pulse duration, frequency, and train duration were held constant.

Many studies have found that increased charge-density at the electrode tip is associated with increased histological damage (Brown et al., 1977; Yuen et al., 1981; Agnew et al., 1983; Bullara et al., 1983; McCreery et al., 1990). Using surface stimulation of cat parietal cortex, McCreery et al. (1990) showed that decreasing the exposed surface area of electrodes from 10 to 1 mm^2 increased the extent of histological damage at the electrode after delivering a train of biphasic pulses (1000 nC/phase) for 7 h. Shrunken tissue and degenerated neurons were observed below the 1.0 mm^2 electrode. The charge density generated at the 1 mm^2 electrode was 1000 nC/mm^2/phase. When an extremely small-tipped electrode (i.e., 0.0065 mm^2) was used to stimulate tissue within the cortex, however, charge densities up to 16000 nC/mm^2/phase did not cause histological damage.

Others have also found that the charge-density threshold for producing histological damage was lower when surface electrodes were used (Bernstein et al., 1977; Pudenz et al., 1975, 1977; Yuen et al., 1981; Robblee et al., 1983), than when depth electrodes were used (Agnew et al., 1986; McCreery et al., 1986). The charge-density threshold was between 300 and 1000 nC/mm^2/phase for pulse trains delivered for many hours at 20–50 Hz when using surface electrodes of 1.0 mm^2. The charge-density threshold was equal to or greater than 16000 nC/mm^2/phase for pulses delivered at comparable train durations and pulse frequencies when using depth electrodes of 0.002 mm^2.

It is noteworthy that the surface electrodes were always made from platinum, whereas the depth electrodes were made from iridium or platinum combined with 30% iridium. It is known that platinum dissolves more readily in neural tissue than does iridium when current densities greater than 1000 nC/mm^2/phase are used (Robblee et al., 1983; Agnew et al., 1986; Tivol et al., 1987), but no simple relationship has been found between platinum dissolution and neuronal damage (Robblee et al., 1983).

The focus of this report is depth stimulation. No study to my knowledge has manipulated electrode tip size systematically for depth stimulation. Therefore, the charge-density threshold for histological damage remains to be determined for depth electrodes having different tip sizes. Using depth electrodes, charge densities as high as 70000 nC/mm^2/phase generated through 0.0001 mm^2 electrodes (e.g., Mitz and Wise, 1987) and as low as 300 nC/mm^2/phase or lower generated through 0.6 mm^2 electrodes (e.g., Yeomans et al., 1985) have been used for evoking behavioral responses.

It has been reported that the charge-density threshold for functional damage is much lower than the threshold for histological damage. McCreery et al. (1986) found that although there was no visible histological damage after stimulating the cortex of cats for 24 h at 20 Hz with charge-balanced biphasic pulses of 32000 nC/mm^2/phase, pyramidal-tract evoked potentials could no longer be elicited from the stimulated sites after such stimulation. Even though histological damage is one way of assessing the noxious effects of brain stimulation, the critical test for such effects is whether the stimulation threshold to evoke a response is stable over time (Yeomans, 1990).

5.4. Electrode materials

Platinum, platinum/iridium, tungsten, or stainless-steel electrodes are often used in brain stimulation. Platinum is used for chronic stimulation since it is a relatively non-toxic to the brain when compared with metals such as gold or rhodium (Bernstein et al., 1977). For depth stimulation, iridium is combined with platinum to give added strength to the electrode. Iridium, although less malleable than platinum, is much more stable for chronic stimulation (Robblee et al., 1983; Agnew et al., 1986; Tivol et al., 1987). Finally, tungsten and stainless-steel are also used for depth stimulation since they exhibit both durability and

strength (Hubel, 1957; Asanuma, 1981; Lemon, 1983; Yeomans, 1990). Materials such as gold, rhodium, or carbon should be avoided (Bernstein et al., 1977).

6. Discussion

Since the time of Ferrier (1886), electrical stimulation has been used to study behaviors subserved by the nervous system (Sherrington, 1906; Lucas, 1913; Adrian, 1921; Penfield and Boldrey, 1937; Hess, 1956; Deutsch, 1964; Wagman, 1964; Ranck, 1975; Yeomans, 1990). Present day stimulation devices activate neurons simultaneously at pre-set currents, pulse durations, frequencies, and train durations. Anyone who has conducted unit recordings will attest to the fact that neurons firing in conjunction with behavioral responses do not all fire simultaneously, nor do they all burst at a uniform rate and duration. For this reason electrical stimulation should not be used in isolation when assessing brain function, because to do so assumes an equivalency between the neural activity that mediates a behavioral response and the electrical stimulation that elicits that response.

This report develops numerous points regarding electrical stimulation. First, current activates neurons directly according to the square of the distance between the electrode tip and the neuron, and the excitability of neurons (i.e., the current-distance constant) can vary between 100 and 4000 $\mu A/mm^2$ using a 0.2-ms cathodal pulse duration. This has been confirmed using both electrophysiological and behavioral methods. To estimate effective current spread, one must take into account the duration of the current pulse. The longer the duration of the pulse, the greater the effective spread of current until further increases in pulse duration no longer increases the effective spread. For all stimulation experiments, therefore, strength-duration functions (e.g.,Fig. 5) should be determined so that current thresholds at different pulse durations may be compared since investigators have been known to use a range of pulse durations (0.1–2.0 ms) to evoke behavioral responses.

Stimulating neurons at their chronaxie readily activates neurons (Lapique, 1905). Furthermore, extremely long-duration pulses (e.g., 2–10 ms) generate multiple action potentials per single pulse (Matthews, 1978; Yeomans et al., 1985; Shizgal et al., 1991) and such pulses are known to increase the refractory period of neurons (Shizgal et al., 1991). The chronaxies for myelinated neurons determined with metal electrodes typically vary from 0.05 to 0.6 ms (Stoney et al., 1968; Ranck, 1975; Asanuma et al., 1976; West and Wolstencroft, 1983; Hentall et al., 1984b). The chronaxie of unmyelinated neurons can be as high as 4 ms (West and Wolstencroft, 1983).

Second, the number of neurons activated directly within a volume of tissue increases monotonically with current. In the cerebral cortex, for instance, the number of neurons activated directly increases according to the volume of the current field. In the cat motor cortex a 10 μA current (at 0.2-ms duration) was estimated to activate 2175 low- (272 $\mu A/mm^2$) and 38 high-threshold neurons (3460 $\mu A/mm^2$). The notion that low currents recruit a few neurons only is clearly wrong, especially if low-threshold neurons contribute disproportionately to an evoked behavioral response. Studies using electrical stimulation to evoke behaviors such as saccades, head turns, or lateral head and body movements have used currents that far exceed 10 μA (e.g., Robinson and Fuchs, 1969; Stryker and Schiller, 1975; Schiller, 1977; Yeomans et al., 1985; McIlwain, 1988; Tehovnik and Lee, 1993). Accordingly, more than a few neurons are involved in generating these behavioral responses.

Since the number of neurons activated directly increases with current, one should maintain a constant level of current when studying behavioral responses to increase the chances that a uniform number of neurons is being activated when comparing different stimulation sites. It is more often the exception (e.g., Tehovnik and Lee, 1993; Tehovnik et al., 1994) than the rule for investigators to hold current constant when mapping the brain with stimulation, because investigators often try to reduce their currents to a minimum to establish a compatibility between behavioral correlates related to unit recording and electrical stimulation. For example, Bruce et al. (1985) showed that currents as low as 50 μA (at 0.25-ms durations) could be used to illustrate a correspondence between the unit activity of a frontal eye field cell that was best modulated by a particular direction and amplitude of saccade and the electrical stimulation through the recording electrode that evoked a saccade of similar direction and amplitude. A current of 50 μA should recruit between 400 and 20 000 neurons directly depending on the distribution of current-distance constants of neurons residing in the frontal eye fields. [2] For over 50% of sites in the frontal eye fields, currents of over 150 μA were required to evoke saccades which means that the number of neurons activated directly in these cases ranged between 2200 and 103 000 neurons. Therefore, in such experiments different numbers of neurons are always being activated at different stimulation sites. If possible, current should be held constant when comparing different sites.

The number of neurons activated indirectly during cortical stimulation is not known. The extent of indirect activation of neurons by different parameters of stimulation needs to be compared for different cortical areas mediating similar behavioral responses (Asanuma and Rosen, 1973;

[2] The estimate assumes a neuronal packing density of 60 cells/0.001^3 (Rockel et al., 1980) for a 2-mm-thick frontal eye field cortex (Winters et al., 1969) and a distribution of current–distance constants ranging from 272 to 3460 $\mu A/mm^2$ for neurons (Stoney et al., 1968).

Jankowska et al., 1975; Asanuma et al., 1976; McIlwain, 1982). That currents over 200 μA were required to evoke saccades from the visual cortex, and that currents as low as 30 μA were sufficient to evoke saccades from the frontal eye fields using comparable conditions (Schiller, 1977) is likely not only indicative of differences in neural excitability, but is also likely related to intrinsic as well as extrinsic synaptic differences between the regions. The distance of synaptic action is dependent on the strength of synapses and on the connectivity of the directly stimulated site. Recently, methods have been developed to determine whether synapses intervene two stimulation sites mediating the same behavior (Yeomans, 1995). Such methods might be useful in studying transynaptic effects within cortex.

Third, the size of an electrode does not affect the current density — nor the effective current spread — at a distance from the electrode. Next to the electrode, however, a small tip generates a higher density of current than does a large tip. A small-tipped electrode not only displaces less tissue as it is lowered into the brain, but it also recruits more high-threshold neurons immediately next to the electrode tip due to the higher current densities generated. A small-tipped electrode should especially be used when one needs to activate high-threshold neurons using low currents.

Fourth, to minimize neuronal damage at the electrode tip, electrode polarization effects need to be controlled. Two methods are typically used to control for polarization: (1) deliver charge-balanced biphasis pulses (e.g., one cathodal pulse followed by an anodal pulse), or (2) electronically discharge the electrodes after the delivery of each monophasic pulse. Also pulse durations that are at or less than the chronaxie of the stimulated tissue should be used to minimize damage.

Finally, it should be added that the evocation of different types of motor behavioral responses requires different amounts of current. Most experiments conducted by Asanuma and colleagues to evoke muscle contractions by stimulating anesthetized cat and monkey cortex typically used currents below 40 μA (at 0.2-ms duration) (Asanuma and Sakata, 1967; Asanuma and Ward, 1971; Asanuma and Rosen, 1972; Asanuma and Arnold, 1975). Others using the non-anesthetized preparation have also found that low currents (< 50 μA at 0.2-ms duration) were sufficient for evoking muscle contractions from cortex (Macpherson et al., 1982; Tanji and Kurata, 1982; Alexander and Crutcher, 1990; Rizzolatti et al., 1990; Luppino et al., 1991). To evoke eye movements from the cortex of non-anesthetized cats and monkeys, however, higher currents (< 3000 μA at 0.2-ms durations) were typically required (Robinson and Fuchs, 1969; Schlag and Schlag-Rey, 1970, 1987; Mohler et al., 1973; Schiller, 1977; Guitton and Mandl, 1978; Bruce et al., 1985; Goldberg et al., 1986; McIlwain, 1988; Tehovnik and Lee, 1993). Accordingly, the amount of current used to evoke a behavioral response depends not only on the current thresholds of the neurons being activated but also on the type of behavioral response being evoked.

The current threshold to evoke a behavioral response is likely related to the number and efficacy of synapses between the stimulated neurons and the muscles that mediate the response. In this regard factor such as the duration of the stimulation train and the frequency of current pulses, both of which affect temporal summation, are important in evoking optimal motor responses. For instance, to evoke circling behavior from rat colliculus, long train durations (10 s) and low frequencies (20–50 Hz) were used (Tehovnik and Yeomans, 1986); whereas to evoke eye movements from rat colliculus, short train durations (0.04–0.1 s) and high frequencies (200 Hz) were used with comparable currents (25–1400 μA at 0.1-ms duration) (McHaffie and Stein, 1982). Thus a range of stimulation parameters should be tested for each new brain area, species, and behavioral response studied to optimize the evoked responses.

In conclusions, (1) current activates a neuron directly according to the square of the distance between the electrode and neuron, (2) even low currents such as 10 μA activate more than a few neurons directly, (3) the electrode tip size has no effect on the effective current spread at far field, (4) to minimize neuronal damage, the electrode should be discharge after each pulse and the pulse duration should not exceed the chronaxie of the stimulated tissue, and (5) when studying a new brain area or behavioral response with electrical stimulation, parametric tests should be done routinely to optimize the evoked response.

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